Object detection using DOTA dataset and training using YOLOV8

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1 Dataset Preparation and Conversion

- Downloaded **DOTA-v1.0** dataset and extracted image and labelTxt.
- Selected **40 images** with aeroplane using a Python script.
- Used **dotadevkit** to convert the dataset to **COCO format** → generated DOTA_1.0.json.
- Split the dataset:
  - 30 images to images/train
  - 10 images to images/validation
- Created corresponding **COCO-style JSON** files: instances_train.json, instances_val.json

2. Folder Structure and Configuration

Organized dataset as:

mini_train/

├── images/

| ├── train/

| └── val/

├── labels/

| ├── train/

| └── val/

├── annotations/

| ├── instances_train.json

| └── instances_val.json

Created a data.yaml file

3. YOLOv8 Setup, & Training

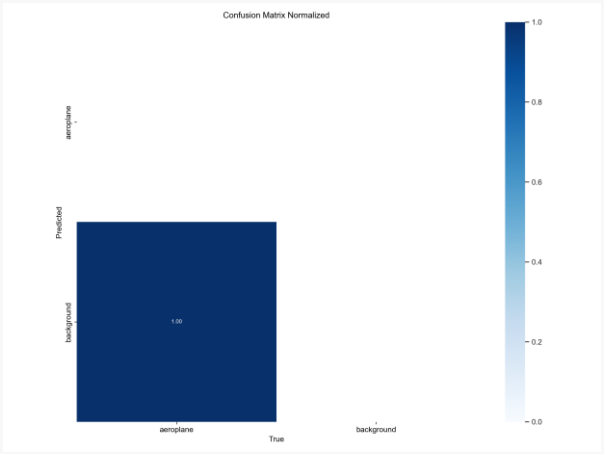
- Installed **Python 3.9** and YOLOv8 using pip install ultralytics.
- Resolved:
  - CUDA compatibility (GPU verified)
- Fixed class ID issues in label .txt files (converted all class IDs to 0).
- Resized all training/validation images to **640×640**.

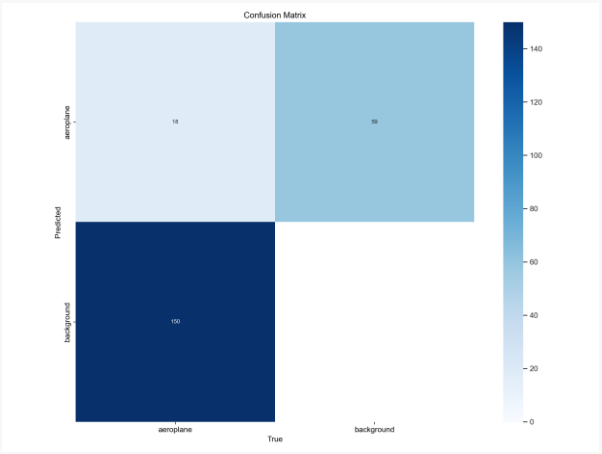
4. Training Models

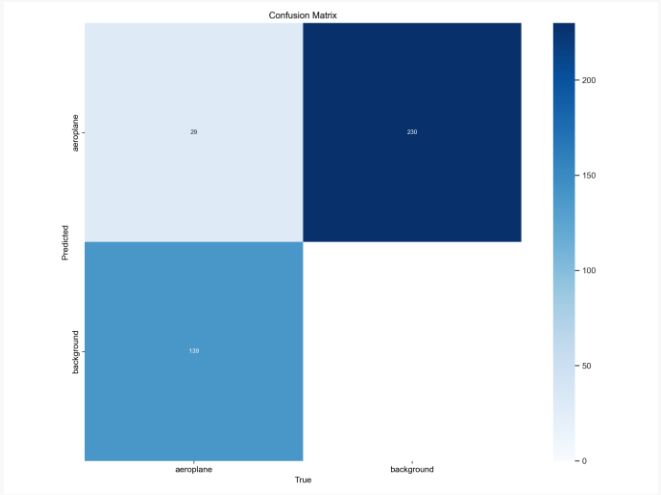
Epoch10(none)

Epoch30(11)

Epoch50(29)





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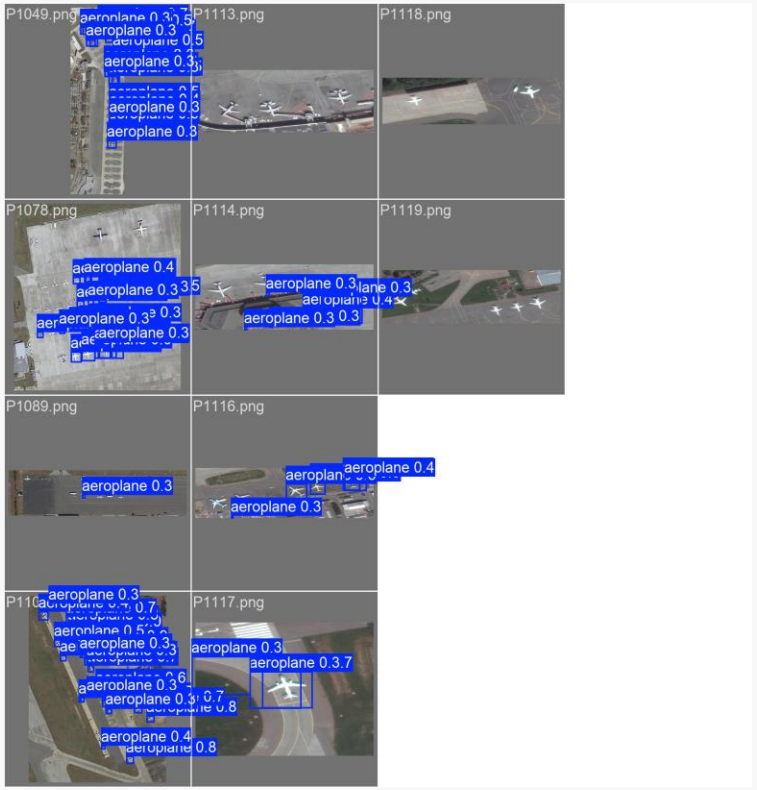
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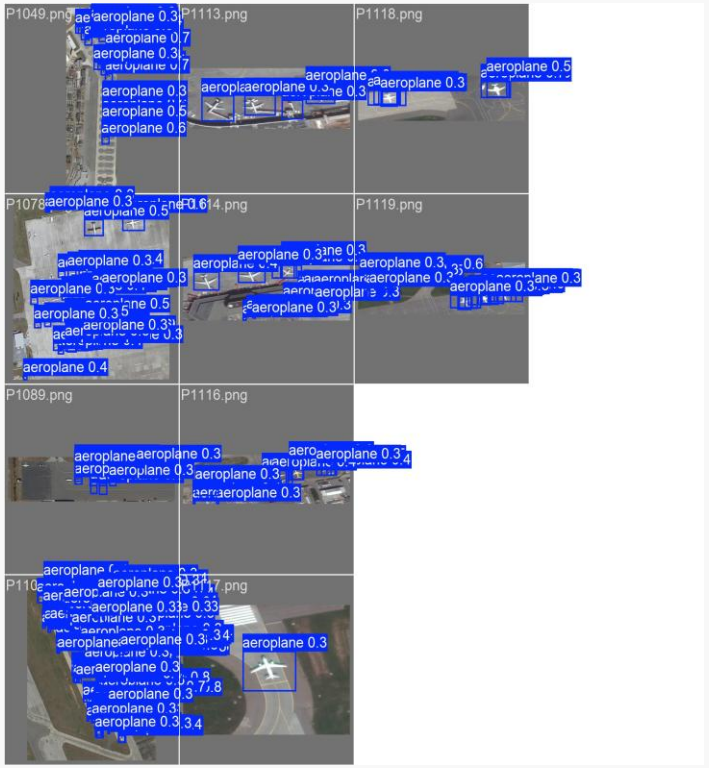
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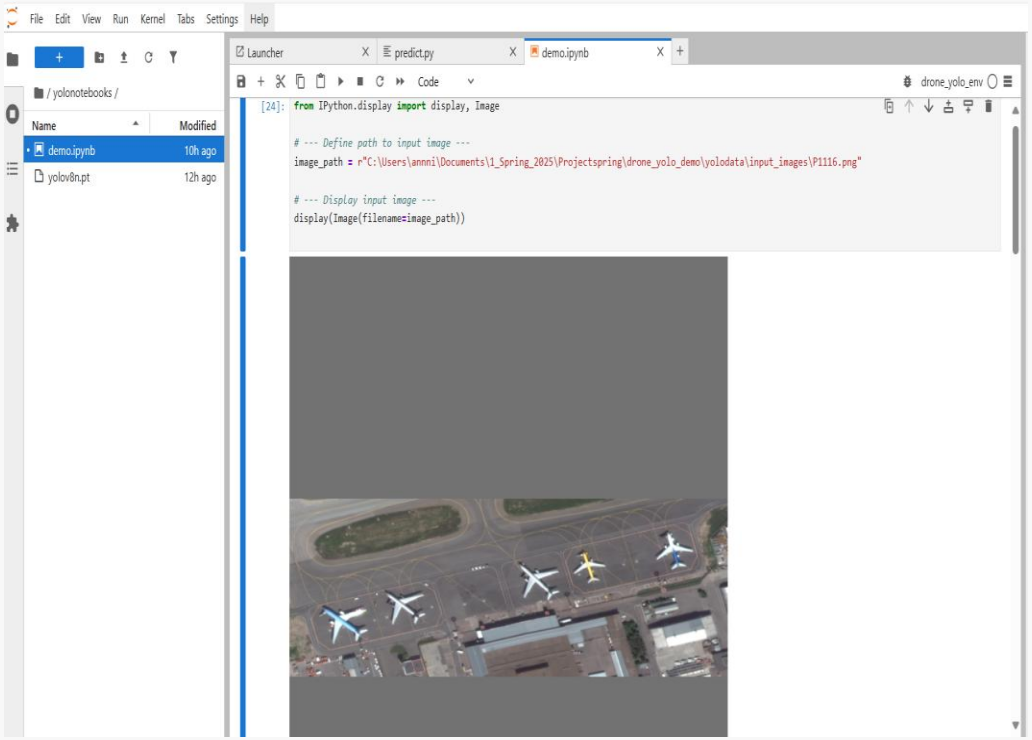
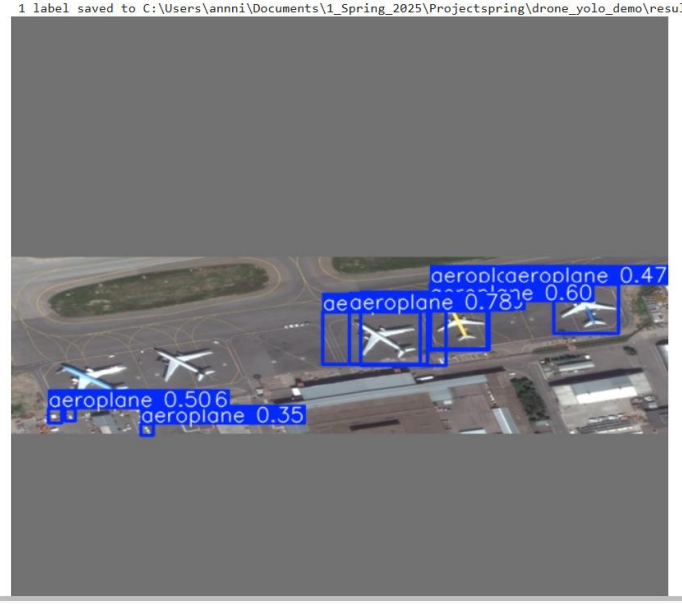
Setting Up a Jupyter environment

- Conda environment named drone_yolo_env
- Necessary packages ultralytics, jupyterlab, and ipython installed
- JupyterLab launched
- A notebook named demo.ipynb used to load and display input drone images using the IPython.display.Image function.
- Using the Epoch30 YoloV8s(best.pt)file performed object detection on a specified input image with a confidence threshold of 0.3.
- Saved to results folder (predict_output2). If the output folder already exists, it is removed to ensure a clean run.
- Displays it using Jupyter's IPython.display.Image function.




```
[30]:
import os
import shutil
from ultralytics import YOLO
from IPython.display import display, Image

# --- Define paths ---
model_path = r"C:\Users\annni\Documents\1_Spring_2025\Projectspring\drone_yolo_demo\weights\best.pt"
output_dir = r"C:\Users\annni\Documents\1_Spring_2025\Projectspring\drone_yolo_demo\results"
predict_name = "predict_output2"
predict_folder = os.path.join(output_dir, predict_name)

# --- Remove old output folder if it exists ---
if os.path.exists(predict_folder):
    shutil.rmtree(predict_folder)

# --- Load YOLOv8 model ---
model = YOLO(model_path)

# --- Run prediction ---
results = model.predict(
    source=image_path,
    save=True,
    save_txt=True,
    project=output_dir,
    name=predict_name,
    imgsz=640,
    conf=0.6
)

# --- Display prediction image ---
predicted_path = os.path.join(results[0].save_dir, os.path.splitext(os.path.basename(image_path))[0] + ".jpg")
display(Image(filename=str(predicted_path)))
```

```
Speed: 9.8ms preprocess, 1196.9ms inference, 4.1ms postprocess per image at shape (1, 3, 640, 640)
Results saved to C:\Users\annni\Documents\1_Spring_2025\Projectspring\drone_yolo_demo\results\predict_output2
1 label saved to C:\Users\annni\Documents\1_Spring_2025\Projectspring\drone_yolo_demo\results\predict_output2\labels
```



Improvements

- Add more imageries to train
- Here worked with YoloV8s try out with Yolov8m
- More Epochs
- Work around with confidence threshold
- Try finding the coordinate locations and overlaying it on a map